



Baseline mathematical aptitude attenuates group differences in posterior P300 during arithmetic verification in high school athletes

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Abstract

Whether the cognitive benefits of athletic training transfer to specific academic domains, such as arithmetic, remains poorly understood. This study investigated the behavioral and electrophysiological (ERP) correlates of multiplication verification in 19 high school cycling athletes and 19 matched controls. Behaviorally, athletes exhibited significantly lower baseline mathematical proficiency (SATM scores), lower accuracy, slower reaction times and poorer processing efficiency. Electrophysiologically, athletes initially demonstrated significantly larger posterior P300 amplitudes than controls. Crucially, these group differences were substantially attenuated after statistically controlling for baseline mathematical proficiency using analysis of covariance (ANCOVA). The findings suggest that arithmetic-related neural processing differences in athletes may be associated with variability in mathematical proficiency rather than reflecting a generalized neurocognitive disadvantage. More broadly, the results support the view that cognitive adaptations associated with athletic expertise may be relatively domain-specific and may not automatically generalize to highly practiced academic skills such as arithmetic verification.

Keywords Cycling athletes · Arithmetic processing · P300 · Event-related potentials (ERP) · Cognitive transfer

Introduction

Extensive research over the past two decades has documented associations between long-term engagement in physical activity and sports participation and neurocognitive functioning (Colcombe and Kramer 2003; Erickson et al. 2015; Hillman et al. 2008; Tan et al. 2025). A substantial body of literature further suggests that the physiological and cognitive demands associated with prolonged athletic training may contribute to experience-dependent neuroplastic changes (Voss et al. 2010; Yao et al. 2024). Athletes have frequently been reported to demonstrate advantages in several domain-general cognitive functions, including attentional control, visuospatial processing, and inhibitory control (Gutiérrez-Capote et al. 2024; Hernández-Mendo et al. 2019; Meng et al. 2019). However, whether these cognitive advantages generalize to domain-specific academic skills remains unclear.

A recent meta-analysis reported that higher skilled athletes performed better than lower skilled athletes on cognitive function tests (Kalén et al. 2021). Likewise, systematic evidence suggests that physical activity and physical fitness are generally associated with better cognitive performance,

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whereas findings regarding academic achievement have been less consistent across studies (Donnelly et al. 2016). In contrast, lower overall academic performance has been observed among young elite athletes compared with their non-elite peers (Pinto-Escalona et al. 2022). These findings indicate that the relationship between cognitive advantages associated with athletic expertise and academic performance remains incompletely understood. Arithmetic fact retrieval is a domain-specific academic skill well suited to addressing this question, because multiplication facts become increasingly represented in semantic memory through years of formal mathematics instruction and repeated practice (Ashcraft 1992; Ericsson et al. 1993). As arithmetic experience accumulates, these facts can increasingly be retrieved directly from semantic memory rather than computed procedurally (Dehaene et al. 2003; Jost et al. 2004).

The cognitive profile of student-athletes cannot be fully understood without acknowledging the educational environments shaping their neurodevelopment. In many intensive youth sports systems, elite athletes face substantial academic time constraints (Pinto-Escalona et al. 2022). Intensive training schedules may constrain the time available for academic activities. However, because the present study did not directly assess study time, arithmetic practice, or educational support, reduced practice should be considered a plausible contextual explanation rather than an established mechanism. Consequently, student-athletes may exhibit differences in arithmetic processing efficiency during arithmetic verification tasks, potentially leading to differences in neural resource allocation during task performance.

The Neural Efficiency Hypothesis provides a theoretical framework for understanding potential differences in neural processing during arithmetic verification (Neubauer and Fink 2009). The hypothesis postulates that experts exhibit lower and more spatially focused cortical activation during domain-specific tasks due to highly optimized neural processing (Babiloni et al. 2010; Dunst et al. 2014). Conversely, when individuals face demanding tasks outside their domain of expertise, they may exhibit greater neural engagement and broader recruitment of attentional resources (Mann et al. 2016; Neubauer and Fink 2009). When task processing is less efficient or requires greater cognitive engagement, individuals may recruit additional executive and attentional resources during task performance. Event-related potentials offer precise temporal resolution to track these resource allocation processes. Among ERP components, the P300 component, particularly the posterior P3b subcomponent, has been widely associated with attentional resource allocation, stimulus evaluation, and context updating (Donchin 1981; Luck 2014; Polich 2007). In particular, P300 amplitude is commonly interpreted as reflecting the allocation of processing resources during stimulus evaluation (Polich

2007). Given that the present study focused on differences in task-related neural engagement and resource allocation during arithmetic verification, P300 amplitude was selected as the primary electrophysiological measure. Arithmetic verification has been widely investigated using ERP paradigms across both child and adult populations, with posterior P300 responses consistently reported during arithmetic evaluation processes (Moore et al. 2014; Cerda et al. 2022, 2023). In arithmetic verification paradigms, P300 responses have been associated with the evaluation and categorization of arithmetic solutions. Correct solutions have been shown to elicit P300 responses in adults when arithmetic problems are presented in familiar digit formats (Dickson et al. 2018). Similarly, adults have been shown to elicit P300 responses to correct multiplication solutions, suggesting that correct answers may be treated as expected targets in highly practiced arithmetic expressions (Grenier et al. 2020). Moreover, larger P300 amplitudes have been observed for small and correct solutions, which are more easily categorized during solution evaluation (Dickson and Wicha 2019). During response evaluation, posterior P300 amplitudes have also been shown to decrease with increasing arithmetic task complexity (Wang et al. 2025). Together, these findings indicate that P300 responses are sensitive to the ease and demands of arithmetic solution evaluation. Accordingly, examining P300 mean amplitudes during multiplication verification may provide insight into evaluative processes involved in arithmetic verification.

To elucidate the boundary conditions of athletic cognitive transfer, the present study examined behavioral performance and P300 mean amplitude differences during an arithmetic verification task between high school cycling athletes and matched controls. Baseline mathematical proficiency was quantified using the Scholastic Assessment Test for Mathematics. Previous research has shown that elite athletes often experience challenges balancing intensive training with academic demands because of time constraints (Gavala-González et al. 2019), and lower overall academic performance has been observed among young elite athletes than their non-elite peers (Pinto-Escalona et al. 2022). Together, these findings suggest that student-athletes may have fewer opportunities for the sustained academic practice required to develop mathematical proficiency. Accordingly, we hypothesized that athletes would exhibit lower mathematical proficiency and reduced behavioral efficiency than matched controls. Furthermore, given the sensitivity of the P300 to arithmetic solution evaluation, we hypothesized that athletes and controls would differ in P300 amplitudes during arithmetic verification. Finally, we examined whether group differences in P300 amplitude remained after statistically accounting for baseline mathematical proficiency.

Methods

Participants

A total of 178 high school students from Hsinchu, Taiwan, were screened, from whom 21 cycling athletes from school varsity teams were identified and compared with matched control participants. Groups were matched on age, gender, handedness, and nonverbal intelligence. When an exact Raven's score match was unavailable (5 of 21 pairs), control participants with the closest scores were selected to maintain group balance. An a priori power analysis using G*Power 3.1 indicated that a total sample of 42 participants (21 per group) would be required to detect a large between-group effect (Cohen's $d=0.90$) with 80% power at $\alpha=0.05$. Two participants were excluded because their task accuracy fell more than two standard deviations below the group mean. An a priori power analysis using G*Power 3.1 indicated that 42 participants (21 per group) would be required to detect a large effect (Cohen's $d=0.90$) with 80% power at $\alpha=0.05$. Following these exclusions, the final sample comprised 38 participants (19 per group).

Participants were senior high school students (Grades 10–12), approximately 16–18 years old. Each group comprised 17 males and 2 females, and all participants were right-handed with normal or corrected-to-normal vision and no history of neurological or psychiatric disorders. Nonverbal intelligence was assessed using the Raven's Standard Progressive Matrices (SPM). Raven scores did not differ significantly between athletes ($M=21.42$, $SD=2.65$) and controls ($M=21.74$, $SD=3.05$), $t(36)=-0.34$, $p=.736$, $d=-0.11$. Foundational mathematical ability was measured using the Scholastic Assessment Test for Mathematics (SATM). As hypothesized, athletes ($M=15.58$, $SD=4.90$) scored significantly lower than control participants ($M=25.95$, $SD=6.93$), $t(36)=-5.32$, $p<.001$, $d=-1.73$.

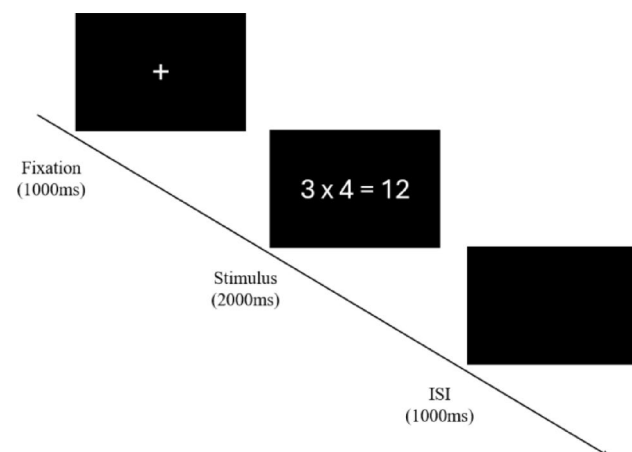


Fig. 1 Illustration of the experimental procedure of the multiplication verification task

Student athletes were recruited from high school varsity cycling teams and engaged in organized training 4–5 times per week, as well as competitive events as part of their athletic programs, meeting commonly used criteria for competitive athletes (Swann et al. 2015). The study was performed in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. The experimental protocol was approved by the Research Ethics Committee of National Tsing Hua University (REC No. 10907HT078). Written informed consent was obtained from all participants and their legal guardians prior to the commencement of the study.

Experimental task and procedure

Participants performed a single-digit multiplication verification task. Operands and solutions were presented simultaneously at the center of the screen (e.g., $3 \times 4 = 12$). Participants judged whether the presented answer was correct or incorrect and responded by pressing designated keys on a keyboard. Both response accuracy and reaction time were recorded.

Each trial began with a fixation cross (1000 ms), followed by the stimulus display (2000 ms) during which responses were made. A 1000 ms interstimulus interval (ISI) preceded the next trial (Fig. 1).

The experiment was programmed using E-Prime (Psychology Software Tools, Pittsburgh, PA). The formal task consisted of 60 trials, with 50% correct and 50% incorrect equations, presented in a randomized order. Tie problems (e.g., 3×3) and rule-based problems involving the operands 0 and 1 were excluded from the stimulus set. Incorrect equations were constructed using operand-related distractors, with false answers designed as multiples of one of the operands in the presented equation. Practice trials were completed until participants understood the instructions. Participants were instructed to respond as quickly and accurately as possible.

EEG recording and preprocessing

Continuous EEG was recorded from 60 scalp electrodes using a SynAmps RT amplifier (Compumedics Neuroscan), with electrodes positioned according to the international 10–20 system (Jasper 1958). Electrode impedances were maintained below 5 k Ω throughout recording. EEG signals were sampled at 250 Hz and were re-referenced to the average reference during offline preprocessing.

EEG preprocessing followed standard procedures for event-related potential (ERP) analysis (Luck 2014) and was implemented using EEGLAB (Delorme and Makeig 2004) and ERPLAB (Lopez-Calderon and Luck 2014) toolboxes

in MATLAB. The continuous data were band-pass filtered between 0.1 and 30 Hz, segmented into epochs from -200 to 1000 ms relative to stimulus onset, and baseline-corrected using the -200 to 0 ms prestimulus interval. Artifact-contaminated trials were rejected using a moving-window peak-to-peak criterion ($80 \mu\text{V}$ threshold, 200 ms window, 100 ms step size), resulting in the exclusion of approximately 10 – 20% of trials. Only correct-response trials were retained for ERP averaging.

Statistical analysis

Behavioral performance, including accuracy, reaction time (RT), and inverse efficiency scores (IES), together with mean P300 amplitudes, was analyzed using independent samples t -tests and mixed-design ANOVAs. For each participant, the inverse efficiency score (IES) was calculated by dividing the mean reaction time on correct-response trials, expressed in milliseconds, by the proportion of correct responses expressed as a value between 0 and 1 . Thus, IES values were expressed in milliseconds, with higher values indicating lower processing efficiency (Townsend and Ashby 1983). The IES has been widely used in cognitive and arithmetic processing research as a combined index of speed and accuracy (Townsend and Ashby 1982).

To examine whether the between-group difference in P300 amplitude remained after adjustment for baseline mathematical proficiency, P300 mean amplitudes were first averaged across the prespecified posterior ROI (P3, Pz, P4, PO3, POz, and PO4). The unadjusted group difference in posterior ROI mean amplitude was initially tested using a one-way analysis of variance (ANOVA). A univariate analysis of covariance (ANCOVA) was subsequently conducted with Group (athletes vs. controls) as the between-subject factor and SATM score as the continuous covariate.

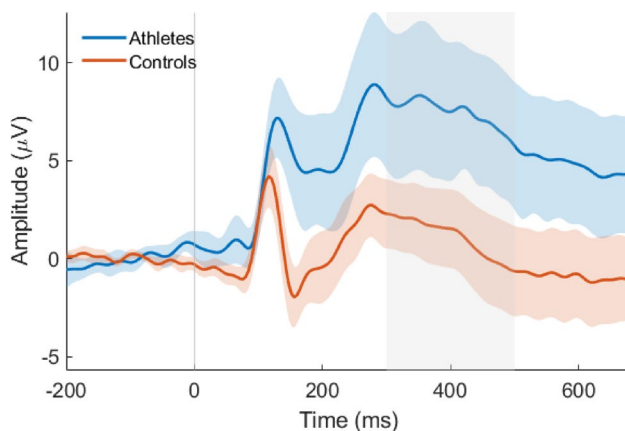


Fig. 2 Grand-average ERP waveforms for athletes and controls averaged across parietal electrodes (P3, Pz, P4, PO3, POz, and PO4). A positive deflection corresponding to the P300 component is observed between 300 – 500 ms. Shaded areas represent 95% confidence intervals

ERP analysis

ERPs were computed by averaging correct-response trials separately for each participant. The P300 component, characterized by a positive deflection over posterior scalp regions, was quantified as the mean amplitude between 300 and 500 ms following stimulus onset (Polich 2007; Luck 2014). P300 mean amplitude was selected as the primary ERP measure because it is commonly interpreted as reflecting attentional resource allocation during task processing (Polich 2007; Wang et al. 2025). To capture the P3b component, analyses focused on parietal electrodes surrounding Pz (P3, Pz, P4, PO3, POz, and PO4), where the P3b typically reaches maximal amplitude (Polich 2007).

To examine scalp distribution, electrodes were grouped into five regions based on the standard 10 – 20 layout: frontal (FP1, FPZ, FP2, AF3, AF4, F7, F5, F3, F1, FZ, F2, F4, F6, F8), central (FC5, FC3, FC1, FCZ, FC2, FC4, FC6, C5, C3, C1, CZ, C2, C4, C6), temporal (FT7, FT8, T7, T8, TP7, TP8), parietal (CP5, CP3, CP1, CPZ, CP2, CP4, CP6, P7, P5, P3, P1, PZ, P2, P4, P6, P8), and occipital (PO7, PO5, PO3, POZ, PO4, PO6, PO8, O1, OZ, O2). Mean amplitudes were averaged within each region and analyzed using a mixed-design ANOVA with Group (athletes vs. controls) as a between-subject factor and Region (frontal, central, temporal, parietal, occipital) as a within-subject factor. Statistical analyses were performed in MATLAB.

Results

Behavioral results

Accuracy was lower in athletes ($M=83.33\%$, $SD=10.54$) than in controls ($M=91.40\%$, $SD=8.70$), $t(36)=-2.57$, $p=.014$, $d=-0.84$. Athletes also showed slower reaction times ($M=1256.21$ ms, $SD=208.98$) than controls ($M=1083.53$ ms, $SD=197.54$), $t(36)=2.62$, $p=.013$, $d=0.85$. Consistently, athletes showed higher inverse efficiency scores ($M=1867$ ms, $SD=626$) than controls ($M=1268$ ms, $SD=316$), $t(36)=3.72$, $p<.001$, $d=1.21$, indicating lower processing efficiency.

ERP results

Grand-average ERP waveforms

As shown in Fig. 2, grand-average ERPs averaged across parietal electrodes (P3, Pz, P4, PO3, POz, and PO4) revealed a prominent positive-going deflection emerging around 300 ms and extending to approximately 500 ms over posterior scalp regions, consistent with the typical P300 component

Fig. 3 Scalp topography of the P300 component (300–500 ms) for athletes and controls. Warmer colors indicate larger positive amplitudes. The P300 shows a posterior distribution, with stronger positivity over parietal and occipital regions in athletes compared with controls

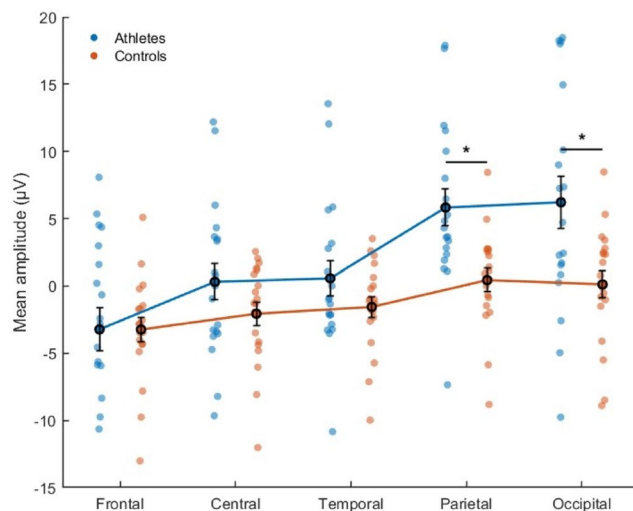
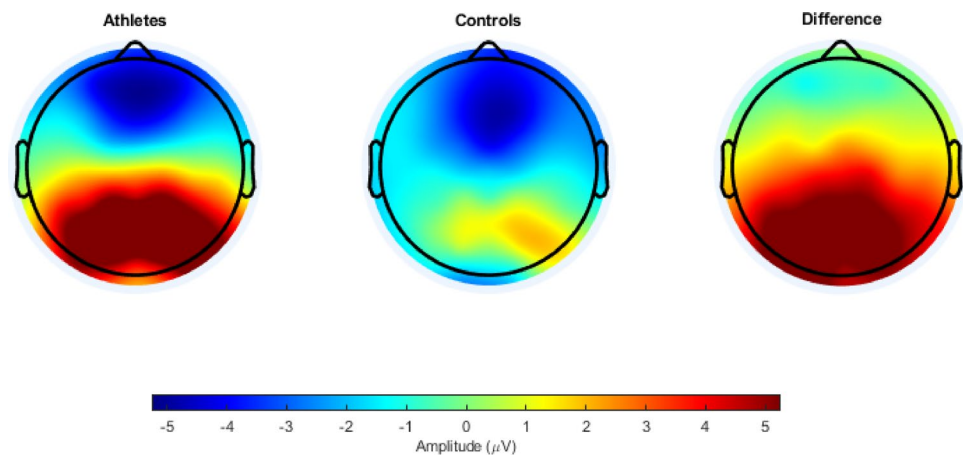


Fig. 4 Mean P300 amplitudes (300–500 ms) across scalp regions (frontal, central, parietal, temporal, and occipital). Dots represent individual participants and error bars indicate standard errors. Significant group differences were observed over parietal ($p = .002$) and occipital ($p = .008$) regions

(Polich 2007; Luck 2014). Within this time window, athletes exhibited larger positive amplitudes than controls.

Scalp topography of the P300

The scalp distribution of mean amplitudes within the 300–500 ms time window is illustrated in Fig. 3. Both groups showed a posteriorly distributed positive potential consistent with the typical P3b component, with maximal amplitudes over parietal and occipital regions. Notably, athletes exhibited stronger posterior positivity compared with controls, particularly over parietal–occipital electrodes.

P300 component

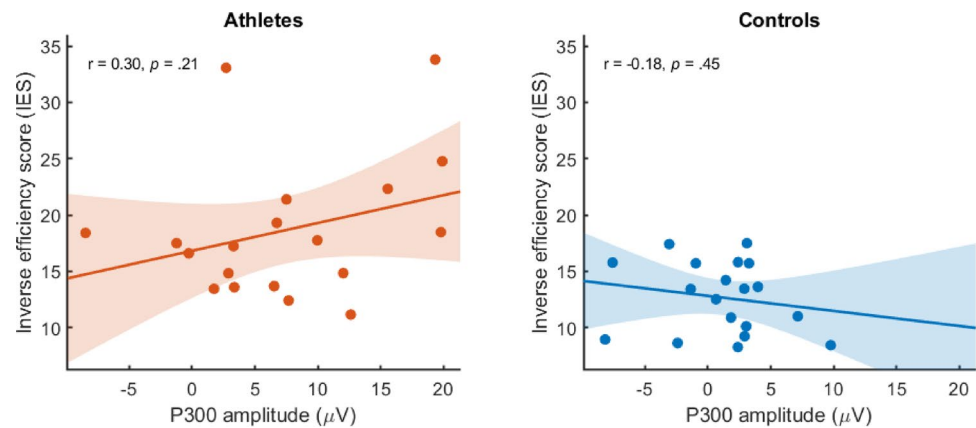
Mean amplitudes within the 300–500 ms time window were analyzed using a 2×5 mixed-design ANOVA with

Group (athletes vs. controls) as a between-subject factor and Region (frontal, central, temporal, parietal, occipital) as a within-subject factor. Mauchly's test indicated that the assumption of sphericity was violated, $W = 0.007$, $\chi^2(9) = 173.22$, $p < .001$; therefore, Greenhouse–Geisser corrections were applied. The analysis revealed a significant main effect of Region, $F(1.29, 46.32) = 12.27$, $p < .001$, $\eta^2 = 0.25$, indicating that P300 amplitudes varied across scalp regions. The main effect of Group was also significant, $F(1, 36) = 5.04$, $p = .031$, $\eta^2 = 0.12$. Importantly, the Group $\times$ Region interaction was significant, $F(1.29, 46.32) = 4.54$, $p = .030$, $\eta^2 = 0.11$.

The scalp distribution of mean amplitudes across regions is shown in Fig. 4. The positivity was most pronounced over posterior scalp regions, particularly parietal and occipital electrodes, indicating a posteriorly distributed P300. Consistent with this topographical pattern, follow-up comparisons revealed significantly larger amplitudes in athletes than in controls over parietal, $t(36) = 3.31$, $p = .002$, $d = 1.07$, and occipital regions, $t(36) = 2.79$, $p = .008$, $d = 0.90$, whereas no significant group differences were observed in frontal ($p = .995$), central ($p = .147$), or temporal regions ($p = .171$). Together, these results indicate that group differences in P300 amplitudes were primarily localized over posterior scalp regions.

As a complementary analysis, we calculated a within-subject correctness effect, defined as the posterior P300 mean amplitude elicited by incorrect equations minus that elicited by correct equations. The correctness effect was $-1.82 \mu\text{V}$ ($SD = 12.68$) in athletes and $0.37 \mu\text{V}$ ($SD = 6.19$) in controls. An independent-samples t test indicated that the magnitude of the correctness effect did not differ significantly between groups, $t(36) = -0.68$, $p = .502$, Cohen's $d = -0.22$. Because this correctness effect is computed within participants as a difference between conditions, it is insensitive to between-group baseline offsets, providing a comparison of arithmetic-related processing that is independent of the overall amplitude difference between groups.

Fig. 5 Relationship between mean P300 amplitude and inverse efficiency scores (IES) in athletes (left) and controls (right). Each point represents an individual participant. Solid lines indicate linear regression fits, and shaded areas represent 95% confidence intervals



The role of mathematical proficiency in group differences of P300 amplitude

To examine whether the larger P300 amplitudes observed in athletes were associated with their lower baseline mathematical skills, an ANCOVA was conducted, controlling for SATM scores. Prior to controlling for SATM, a significant main effect of Group on P300 amplitude was observed, $F(1, 36)=9.72$, $p = .004$, $\eta^2 = 0.213$. After adjusting for SATM, the previously significant main effect of Group on P300 amplitude was no longer statistically significant, $F(1, 35)=1.80$, $p = .188$, $\eta^2 = 0.049$. The Group–Region interaction also became non-significant ($p > .10$).

Relationship between P300 amplitude and processing efficiency

Pearson correlations were conducted to examine the relationship between mean P300 amplitudes averaged across parietal electrodes (P3, Pz, P4, PO3, POz, and PO4) and IES, an index of processing efficiency (Fig. 5). No statistically significant correlations were observed between P300 amplitudes and IES in either the athlete group ($r = .30$, $p = .21$) or the control group ($r = -.18$, $p = .45$).

To determine whether the brain–behavior relationships differed between groups, Fisher’s r -to- z tests were conducted. The results indicated that the correlations did not differ significantly between groups, $z = 1.41$, $p = .16$.

Discussion

The present study investigated behavioral performance and posterior P300 activity during a domain-specific arithmetic verification task in high school student-athletes and matched controls. Behaviorally, athletes exhibited lower accuracy, slower reaction times, and higher IES, together with lower baseline mathematical proficiency. Despite

these behavioral differences, athletes showed significantly larger posterior P300 amplitudes. However, the group difference in P300 amplitude was substantially attenuated after statistically controlling for baseline mathematical proficiency. Previous research has reported positive associations between physical activity, physical fitness, and cognitive performance, as well as better cognitive test performance in higher-skilled athletes (Donnelly et al. 2016; Kalén et al. 2021). In comparison, less is known about behavioral and neural differences between athletes and non-athletes during domain-specific academic tasks. The present findings from the arithmetic verification task suggest that athletes exhibited lower behavioral performance and mathematical proficiency than matched controls, while the initially observed P300 group difference was attenuated after accounting for mathematical proficiency. These findings highlight the relevance of domain-specific proficiency when evaluating neurocognitive differences in student-athletes.

The behavioral findings and lower mathematical proficiency scores observed in the athlete group may reflect differences in arithmetic processing efficiency during multiplication verification. Arithmetic fact retrieval is generally considered a highly practiced cognitive process that increasingly relies on semantic memory retrieval with development and repeated scholastic experience (Ashcraft 1992; Dehaene et al. 2003; Jost et al. 2004). In intensive youth sports systems, substantial training demands may reduce the time available for repetitive academic practice (Pinto-Escalona et al. 2022), potentially contributing to variability in arithmetic fluency and mathematical proficiency across individuals.

The present findings are broadly consistent with literature suggesting that cognitive adaptations associated with athletic expertise may be relatively domain-specific and may not automatically generalize to highly practiced academic skills (Kalén et al. 2021; Sala and Gobet 2017). However, the present study did not directly assess arithmetic strategy use, and therefore the observed neural differences should not be interpreted as definitive evidence of reliance on

procedural calculation strategies. In addition, because the operands and solutions were presented simultaneously, the present design did not temporally isolate neural responses to operand processing from those associated with solution evaluation. Previous arithmetic ERP studies using sequential presentation have reported P300 responses time-locked to solution onset (Dickson et al. 2018; Dickson and Wicha 2019; Grenier et al. 2020). Accordingly, the P300 activity observed in the present study cannot be specifically attributed to solution evaluation. This difference in task presentation should be considered when comparing the present findings with solution-locked P300 effects reported in previous arithmetic verification studies. Although the present study focused on P300 amplitude as an index of task-related resource allocation, P300 latency may provide complementary information regarding the temporal aspects of arithmetic solution evaluation and should be examined in future research.

The electrophysiological findings revealed group differences in posterior P300 amplitudes during arithmetic verification. Previous arithmetic ERP studies have shown larger P300 amplitudes for small and correct solutions that were more easily categorized (Dickson and Wicha 2019). During response evaluation, posterior P300 amplitudes have also been shown to decrease with increasing arithmetic task complexity (Wang et al. 2025). Across these studies, larger P300 amplitudes were observed for more easily categorized solutions or under lower task complexity. In the present study, however, athletes exhibited larger posterior P300 amplitudes despite lower behavioral efficiency and mathematical proficiency. Therefore, the larger P300 amplitudes observed in athletes should not be interpreted as evidence of greater arithmetic automaticity or easier solution categorization. Importantly, the group difference in P300 amplitude was attenuated after controlling for mathematical proficiency, indicating that the observed group difference was reduced when between-group variability in mathematical proficiency was statistically accounted for. Exercise-related modulation of norepinephrine-related attentional processes may also contribute to differences in P300 activity, potentially through alterations in tyrosine hydroxylase regulation (Nieuwenhuis et al. 2005; Roelands et al. 2015).

The covariance analysis further suggests that baseline mathematical proficiency may partially account for the observed group differences in posterior P300 amplitudes. After statistically controlling for mathematical proficiency, the group differences in P300 amplitudes were no longer statistically significant. These findings are consistent with the possibility that mathematical proficiency contributes to variability in neural processing during arithmetic verification. This interpretation is broadly compatible with the Neural Efficiency Hypothesis (Neubauer and Fink 2009), which

proposes that differences in domain-specific experience and proficiency may influence the efficiency of neural processing during task performance. Nevertheless, these findings should be interpreted in light of the relatively modest sample size (19 participants per group), which may have limited statistical power to detect small between-group differences. Future studies with larger samples are needed to examine the robustness and generalizability of the present findings.

In conclusion, the present findings suggest that cognitive advantages associated with athletic expertise may not uniformly transfer to highly practiced academic domains such as arithmetic verification. Although athletes have been reported to demonstrate specialized adaptations in certain perceptual-cognitive and executive domains (Yao et al. 2024), the present findings indicate that behavioral and neural differences may still emerge in academic tasks requiring arithmetic evaluation. More broadly, these findings highlight the importance of considering domain-specific educational experiences when interpreting neurocognitive performance in student-athlete populations.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00221-026-07366-y>.

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Author contributions Conceptualization: Chia-Chi Hsu, Zai-Fu Yao, Tsu-Jen Ding, Hui-Yu Hsu; Methodology: Chia-Chi Hsu, Zai-Fu Yao, Chen-Yu Yao; Investigation (Data collection): Tsu-Jen Ding, Hui-Yu Hsu; Data curation: Chia-Chi Hsu, Chen-Yu Yao; Formal analysis: Chia-Chi Hsu, Zai-Fu Yao; Writing - original draft preparation: Chia-Chi Hsu; Writing - review and editing: Zai-Fu Yao; Technical supervision (EEG analysis): Chen-Yu Yao; Supervision: Zai-Fu Yao.

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Data availability The data supporting this study's findings are available from the authors, though access is restricted due to research ethics considerations. These data are not publicly accessible but can be obtained from the authors upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval This study was performed in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. The experimental protocol was approved by the Research Ethics Committee under the number REC No. 10907HT078.

Informed consent Informed consent was obtained from all individual participants and their legal guardians included in the study.

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